Unbalanced Mach–Zehnder Interferometers for Measuring the Group Index of Silicon Strip Waveguides

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***Abstract*—This report describes the design of four unbalanced Mach–Zehnder interferometers (MZIs) and a grating coupler loopback for the openEBL-2026-10 electron beam lithography run. All devices use $500\,\mathrm{nm}$ wide and $220\,\mathrm{nm}$ thick silicon strip waveguides in the fundamental quasi-TE mode near $1550\,\mathrm{nm}$. The MZIs have path length differences of 50, 100, 150 and $200\,\mu\mathrm{m}$. Mode and circuit simulations give a group index of 4.22 at $1550\,\mathrm{nm}$ and free spectral ranges between 2.85 and $11.39\,\mathrm{nm}$, and the group index extracted from the simulated spectra agrees with the waveguide model. The free spectral range of each MZI will be measured and used to calculate the group index of the fabricated waveguide, which will then be compared with simulation.**

***Index Terms*—Silicon photonics, Mach–Zehnder interferometer, group index, compact model.**

I. Introduction

SILICON photonics uses the large refractive index contrast between silicon ($n \approx 3.48$) and silicon dioxide ($n \approx 1.44$) at $1550\,\mathrm{nm}$ to confine light in waveguides that are only a few hundred nanometres wide. This allows compact optical circuits to be made with the same fabrication tools that are used for CMOS electronics [1]. Mach–Zehnder interferometers (MZIs) are one of the most common building blocks in these circuits. They are used as wavelength filters and multiplexers in optical communication links, as switches and modulators, and as sensors.

The group index n_g of a waveguide is needed to design most interferometers and resonators, because it sets their free spectral range (FSR), for example the channel spacing of a wavelength filter. It can be calculated with a mode solver, but the fabricated waveguide is never exactly the designed one, and small changes in its width and thickness change n_g. It is therefore useful to measure n_g on the fabricated chip.

An unbalanced MZI is one of the simplest ways to do this. Its transmission spectrum is periodic in wavelength, and the period depends only on n_g and on the length difference ΔL between the two arms. Since ΔL is fixed by the layout, measuring the FSR gives n_g.

The objective of this design is to extract n_g of the fabricated waveguide from measured MZI spectra and compare it with the simulated value. For this, I chose four MZIs with $\Delta L = 50$, 100, 150 and $200\,\mu\mathrm{m}$, and a loopback structure made of two grating couplers joined by a straight waveguide. The loopback measures the response of the grating couplers, so that it can be removed from the MZI spectra. Having four values of ΔL gives four separate estimates of n_g from the same chip, which can be compared with each other and with the simulation.

The chip is fabricated by Applied Nanotools on an SOI wafer with a $220\,\mathrm{nm}$ silicon layer, using a single full etch and an oxide cladding [2]. The designs have to fit in a $605\,\mu\mathrm{m} \times 410\,\mu\mathrm{m}$ area. Light is coupled into and out of the chip with TE grating couplers spaced by $127\,\mu\mathrm{m}$, the pitch of the fibre array in the measurement setup. According to the course guidelines, the tunable laser has a minimum step of $1\,\mathrm{pm}$ and the grating couplers limit the usable bandwidth to about $50\,\mathrm{nm}$. The FSR of each MZI must therefore be well below $50\,\mathrm{nm}$, so that several fringes fall inside the measured range.

II. Theory

A. Effective index, group index and compact model

The waveguide is used in its fundamental quasi-TE mode, in which the electric field is mostly parallel to the chip surface. The phase of this mode is described by its effective index n_{eff}. Over a length L the mode picks up a phase βL, where $\beta = 2\pi n_{\mathrm{eff}}/\lambda$ is the propagation constant and λ is the free-space wavelength.

In silicon waveguides n_{eff} changes strongly with wavelength, mainly because the size of the mode changes with wavelength (waveguide dispersion). Over a limited wavelength range it can be written as a second-order polynomial around the centre wavelength $\lambda_0 = 1.55\,\mu\mathrm{m}$ [1]:

$$n_{\mathrm{eff}}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \tag{1}$$

where n_1, n_2 and n_3 are found by fitting this expression to mode solver results. Equation (1) is the compact model of the waveguide, and it is used later in the circuit simulations.

The group index gives the speed of a wave packet, $v_g = c/n_g$, and is defined as

$$n_g(\lambda) = n_{\mathrm{eff}}(\lambda) - \lambda \frac{dn_{\mathrm{eff}}}{d\lambda}. \tag{2}$$

At $\lambda = \lambda_0$, using (1), this becomes $n_g(\lambda_0) = n_1 - \lambda_0 n_2$. Because n_{eff} decreases with wavelength ($n_2 < 0$), n_g is larger than n_{eff}.

B. MZI transfer function

In the MZI (Fig. 1), a Y-branch splits the input light equally into two arms of length L_1 and L_2, and a second Y-branch

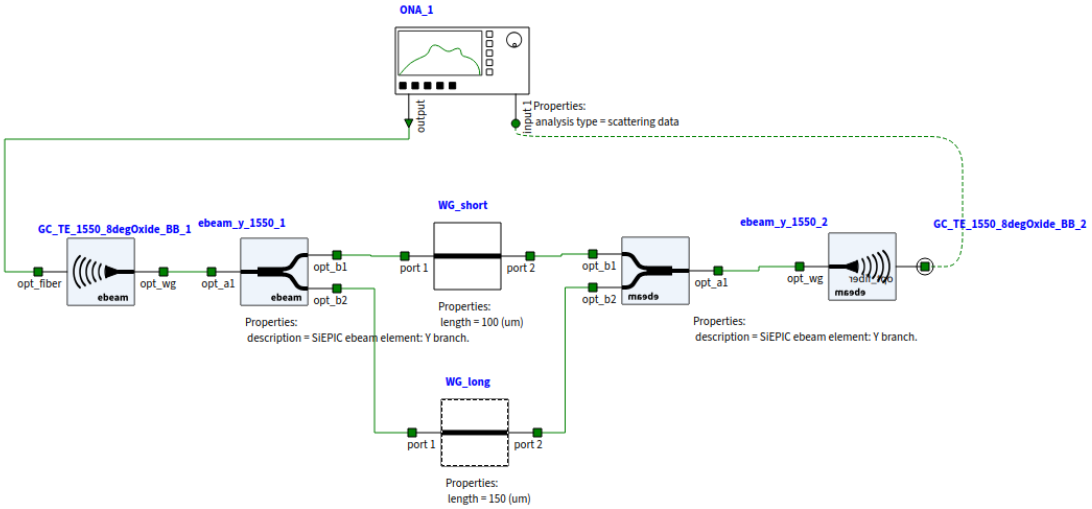


Fig. 1. MZI circuit in Lumerical INTERCONNECT, shown for $\Delta L = 50\,\mu\text{m}$. For the loopback, the Y-branches and waveguides are removed and the two grating couplers are connected directly.

combines the two arms into the output waveguide. Each Y-branch multiplies the field in each arm by $1/\sqrt{2}$, so the output intensity is [1]

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \frac{1}{4}\left|e^{-j\beta L_1 - \alpha L_1/2} + e^{-j\beta L_2 - \alpha L_2/2}\right|^2 \tag{3}$$

where α is the power loss per unit length of the waveguide. If the loss is neglected, (3) becomes

$$\frac{I_{\text{out}}}{I_{\text{in}}} = \frac{1}{2}\left[1 + \cos(\beta\,\Delta L)\right] \tag{4}$$

with $\Delta L = L_2 - L_1$. When the light from the two arms arrives in phase ($\beta\Delta L = 2\pi m$), all of it reaches the output. When it arrives out of phase, the combined field cannot be guided by the output waveguide and is radiated at the second Y-branch, so the transmission drops to zero.

In the measurement, the light also passes through two grating couplers, whose transmission $T_{\text{GC}}(\lambda)$ depends on wavelength. The measured spectrum is therefore $T_{\text{GC}}^2(\lambda)$ multiplied by (3). The loopback structure measures $T_{\text{GC}}^2(\lambda)$ on its own, so it can be subtracted from the MZI spectra in dB.

C. Group index from the free spectral range

The FSR is the wavelength spacing between two neighbouring minima (or maxima) of (4). Between two neighbouring minima at λ_1 and $\lambda_2 = \lambda_1 + \text{FSR}$, the phase difference between the arms changes by 2π:

$$\beta(\lambda_1)\,\Delta L - \beta(\lambda_2)\,\Delta L = 2\pi. \tag{5}$$

Since the FSR is small compared with λ, the change in β can be written using its derivative:

$$\beta(\lambda_1) - \beta(\lambda_2) \approx -\frac{d\beta}{d\lambda}\,\text{FSR}. \tag{6}$$

Differentiating $\beta = 2\pi n_{\text{eff}}/\lambda$ and using (2),

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda}\frac{dn_{\text{eff}}}{d\lambda} - \frac{2\pi n_{\text{eff}}}{\lambda^2} = -\frac{2\pi n_g}{\lambda^2}. \tag{7}$$

Substituting (6) and (7) into (5) gives

$$\text{FSR} = \frac{\lambda^2}{n_g\,\Delta L} \tag{8}$$

and, solving for the group index,

$$n_g = \frac{\lambda^2}{\text{FSR}\cdot\Delta L}. \tag{9}$$

Equation (9) is used to find n_g from both the simulated and the measured spectra, with λ taken as the midpoint between the two minima. It also shows that the FSR is set by the group index and not by the effective index. Using n_{eff} instead would give an FSR that is too large by a factor of n_g/n_{eff}.

III. Modelling and Simulation

A. Waveguide geometry and mode profile

The waveguide is a fully etched silicon strip, $500\,\text{nm}$ wide and $220\,\text{nm}$ thick, with silicon dioxide below and above it, as in the fabrication process [2]. Its cross-section was simulated with the finite-difference eigenmode (FDE) solver in Lumerical MODE [3], using the Palik material data for Si and SiO_2 [4], a $3\,\mu\text{m} \times 2.5\,\mu\text{m}$ simulation region and a $10\,\text{nm}$ mesh.

At $1550\,\text{nm}$ the solver finds the quasi-TE mode with $n_{\text{eff}} = 2.445$ (TE polarization fraction 98%) and the quasi-TM mode with $n_{\text{eff}} = 1.772$. A third mode, the first-order TE mode, has $n_{\text{eff}} = 1.494$. This is only slightly above the index of the oxide (1.444), so this mode is weakly guided. Its field is antisymmetric, so it is not excited by the symmetric grating couplers and Y-branches used in this design. Fig. 2 shows the quasi-TE mode, which is the mode used in all devices.

B. Effective and group index versus wavelength

The mode was solved at 21 wavelengths between 1500 and $1600\,\text{nm}$. Fig. 3 shows the effective index (a) and the group

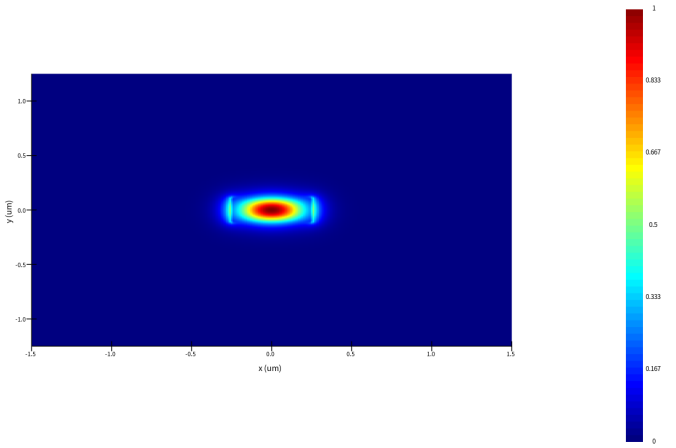


Fig. 2. Electric field intensity $|E|^2$ (normalized) of the quasi-TE mode at $1550\,\mathrm{nm}$, from the Lumerical FDE simulation. The bright lines at the vertical sidewalls come from the jump in the horizontal electric field at the silicon–oxide boundary.

TABLE I
PROPOSED DEVICES AND EXPECTED FSR AT 1550 NM

Device	ΔL (µm)	FSR (nm)	Periods in 50 nm
MZI1	50	11.39	4.4
MZI2	100	5.69	8.8
MZI3	150	3.80	13.2
MZI4	200	2.85	17.6
Loopback	–	–	–

index (b). Fitting (1) to the n_{eff} values gives the compact model

$$n_{\mathrm{eff}}(\lambda) = 2.4453 - 1.1444\,(\lambda-\lambda_0) - 0.4372\,(\lambda-\lambda_0)^2 \quad (10)$$

with λ in µm and $\lambda_0 = 1.55\,\mu\mathrm{m}$. The rms difference between the fit and the FDE results is 1.4×10^{-4}. Using (2), the group index at $1550\,\mathrm{nm}$ is $n_g = 2.4453 + 1.55\times1.1444 = 4.219$.

The group index in Fig. 3(b) was calculated from the FDE n_{eff} values by finite differences, and it shows a step near $1530\,\mathrm{nm}$. This is most likely caused by the tabulated silicon data used by the solver: the index is interpolated linearly between data points, so its slope $dn/d\lambda$ changes suddenly at a data point. The compact model smooths over this step. At exactly $1550\,\mathrm{nm}$ the solver gives $n_g = 4.241$, about 0.5% higher than the value from the fit. Equation (10) is used for all the waveguides in the circuit simulations.

C. Design variations

Table I lists the four MZIs and their expected FSR at $1550\,\mathrm{nm}$, calculated with (8) and $n_g = 4.219$. All FSRs are well below the $50\,\mathrm{nm}$ measurement bandwidth, so each spectrum will contain between about 4 and 18 periods. This gives several FSR values per device, so n_g can be found at several wavelengths. Even the smallest FSR ($2.85\,\mathrm{nm}$) is sampled by more than 2800 points with the $1\,\mathrm{pm}$ laser step. The loopback has no FSR; it is only used to measure the grating coupler response.

TABLE II
GROUP INDEX FROM THE SIMULATED SPECTRA, USING THE TWO MINIMA ON EITHER SIDE OF 1550 NM

ΔL (µm)	λ_1 (nm)	λ_2 (nm)	FSR (nm)	n_g, Eq. (9)
50	1542.970	1554.344	11.374	4.217
100	1545.802	1551.488	5.686	4.218
150	1546.748	1550.539	3.791	4.218
200	1547.221	1550.064	2.843	4.218

D. Circuit simulation

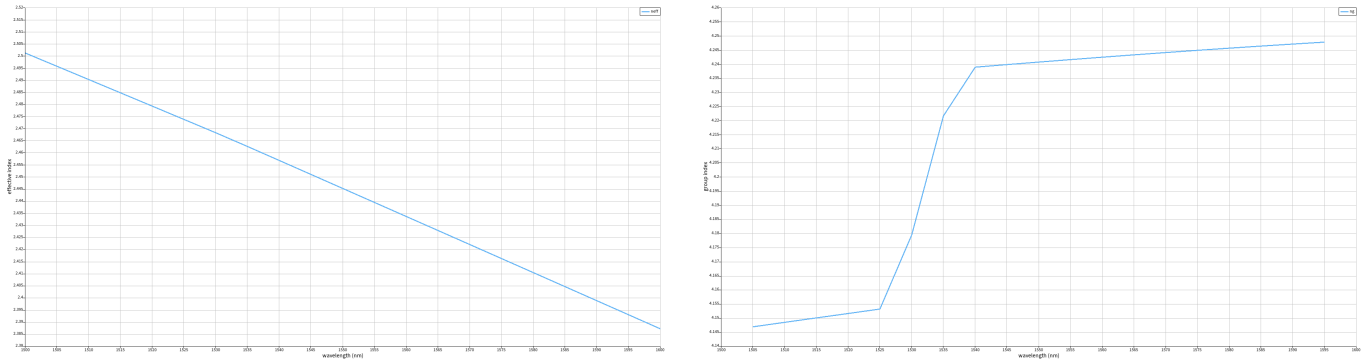
The MZIs were simulated in Lumerical INTERCONNECT [3] with the circuit in Fig. 1. The grating couplers (GC_TE_1550_8degOxide_BB) and the Y-branches (ebeam_y_1550) [5] are compact models from the SiEPIC EBeam library [6], and are the same components that will be used in the layout. The two arms are straight waveguides described by the compact model (10), entered as an effective index of 2.4453, a group index of 4.2192 and a dispersion of 4.52×10^{-3} s/m^2 at $1550\,\mathrm{nm}$, with an assumed loss of $3\,\mathrm{dB/cm}$. For ideal Y-branches, the transmission of each MZI is then given by (3) with β calculated from (10). The short arm is $100\,\mu\mathrm{m}$ long, and the long arm was set to 150, 200, 250 and $300\,\mu\mathrm{m}$ for the four MZIs. The transmission from 1500 to $1600\,\mathrm{nm}$ was calculated with an optical network analyzer using 20001 points. The loopback was simulated in the same way, with the two grating couplers connected directly.

Fig. 4 shows the simulated spectra for $\Delta L = 50$ and $200\,\mu\mathrm{m}$ and for the loopback. The loopback has a maximum transmission of $-4.7\,\mathrm{dB}$ at $1554.5\,\mathrm{nm}$ and a $3\,\mathrm{dB}$ bandwidth of $45\,\mathrm{nm}$, and it forms the envelope of the MZI spectra. The MZI maxima are about $0.3\,\mathrm{dB}$ below the loopback, which is the loss of the two Y-branches and the waveguides. The FSRs measured on the simulated spectra at $1550\,\mathrm{nm}$ agree with Table I to within $0.001\,\mathrm{nm}$.

E. Group index from the simulated spectra

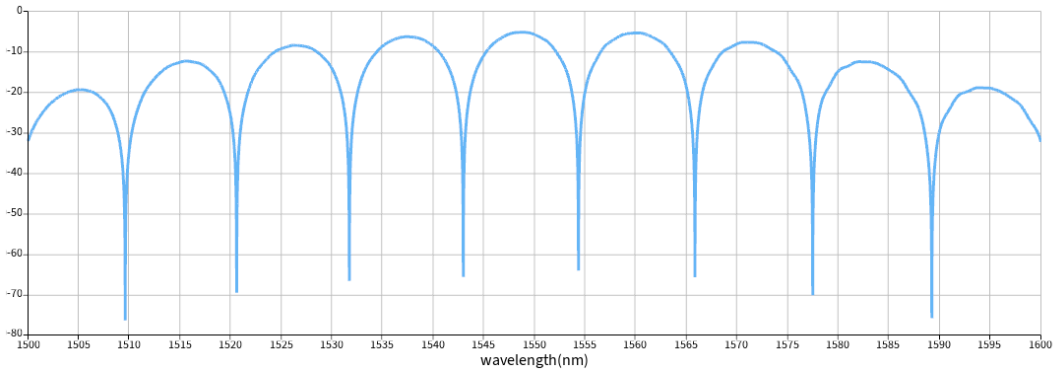
To test (9), the loopback spectrum was subtracted from each MZI spectrum (in dB) to remove the grating coupler response, and the wavelengths of the minima were found. Table II lists, for each MZI, the two minima on either side of $1550\,\mathrm{nm}$, the FSR between them, and the group index from (9) at their midpoint, which is about $1548.65\,\mathrm{nm}$ for all four devices. All four MZIs give $n_g = 4.217$–4.218, the same as the compact model at this wavelength (4.217). Repeating the calculation for every pair of neighbouring minima between 1500 and $1600\,\mathrm{nm}$ gives values that agree with the model to within 0.06%. This confirms that (9) recovers the group index used in the simulation.

With the measured data, n_g will be found in the same way. The measured loopback spectrum will be subtracted to remove the grating coupler response, the minima of each MZI spectrum will be located, and (9) will give n_g at each midpoint wavelength. The four MZIs give four independent sets of values, which will be compared with each other and with the simulation. Any difference from the simulation will mostly come from the fabricated width and thickness of the

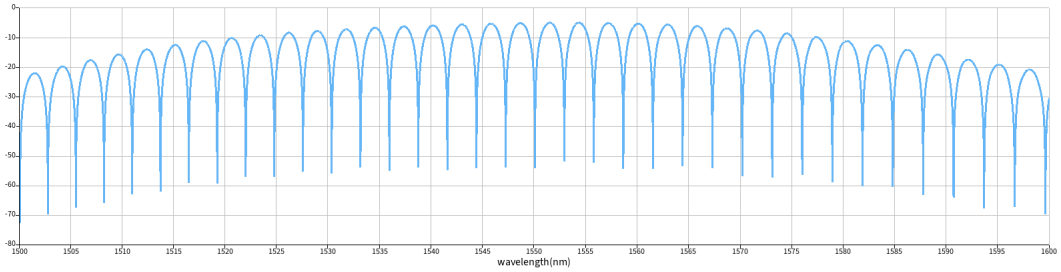


(a) (b)

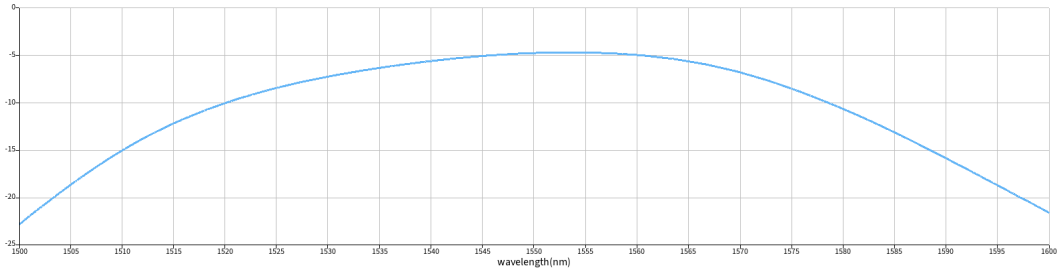
Fig. 3. (a) Effective index and (b) group index of the quasi-TE mode versus wavelength, from the FDE wavelength sweep (plotted in Lumerical). The group index was calculated from the effective index with (2), using the slope between neighbouring points.



(a) MZI, $\Delta L = 50\,\mu\text{m}$



(b) MZI, $\Delta L = 200\,\mu\text{m}$



(c) Loopback

Fig. 4. Simulated transmission in dB, including both grating couplers, versus wavelength (Lumerical INTERCONNECT).

waveguide. As a second method, (3) with the compact model (1) can be fitted to each measured spectrum, using n_1, n_2 and n_3 as fitting parameters.

IV. Fabrication

To be completed.

V. Experimental Data

To be completed.

VI. Analysis

To be completed.

VII. Conclusion

To be completed.

References

[1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design: From Devices to Systems*. Cambridge, U.K.: Cambridge Univ. Press, 2015.
[2] SiEPIC openEBL-2026-10 design submissions. [Online]. Available: https://github.com/SiEPIC/openEBL-2026-10
[3] Ansys Lumerical MODE and INTERCONNECT, 2026 R1. [Online]. Available: https://optics.ansys.com
[4] E. D. Palik, Ed., *Handbook of Optical Constants of Solids*. Orlando, FL, USA: Academic Press, 1985.
[5] Y. Zhang, S. Yang, A. E.-J. Lim, G.-Q. Lo, C. Galland, T. Baehr-Jones, and M. Hochberg, "A compact and low loss Y-junction for submicron silicon waveguide," *Opt. Express*, vol. 21, no. 1, pp. 1310–1316, 2013.
[6] SiEPIC EBeam Process Design Kit. [Online]. Available: https://github.com/SiEPIC/SiEPIC_EBeam_PDK